LES BATTERIES AU PLOMB DANS LE SPORT AUTOMOBILE



TARIQ
ANAS
14919

AVANTAGES	INCONVENIENTS
COÛT	POIDS
ROBUSTESSE	DURÉE DE VIE
RECYCLABILITÉ	PERFORMANCE

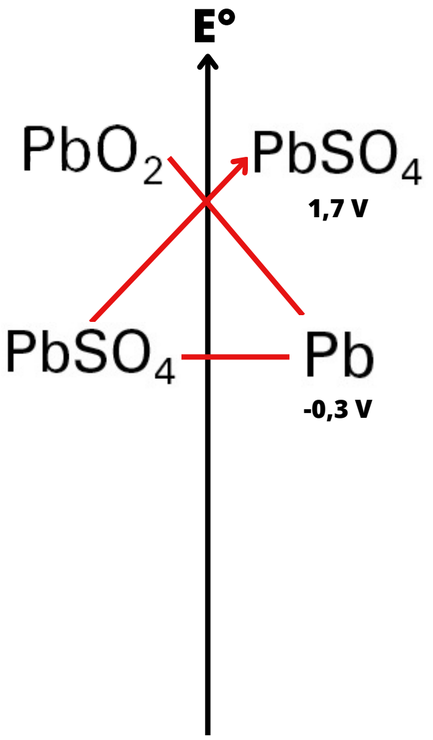
SOMMAIRE

1 MODÈLE EXPÉRIMENTAL

2 INFLUENCE DE LA PUISSANCE DÉLIVRÉE

3 INFLUENCE DE LA TEMPÉRATURE

1 MODÈLE

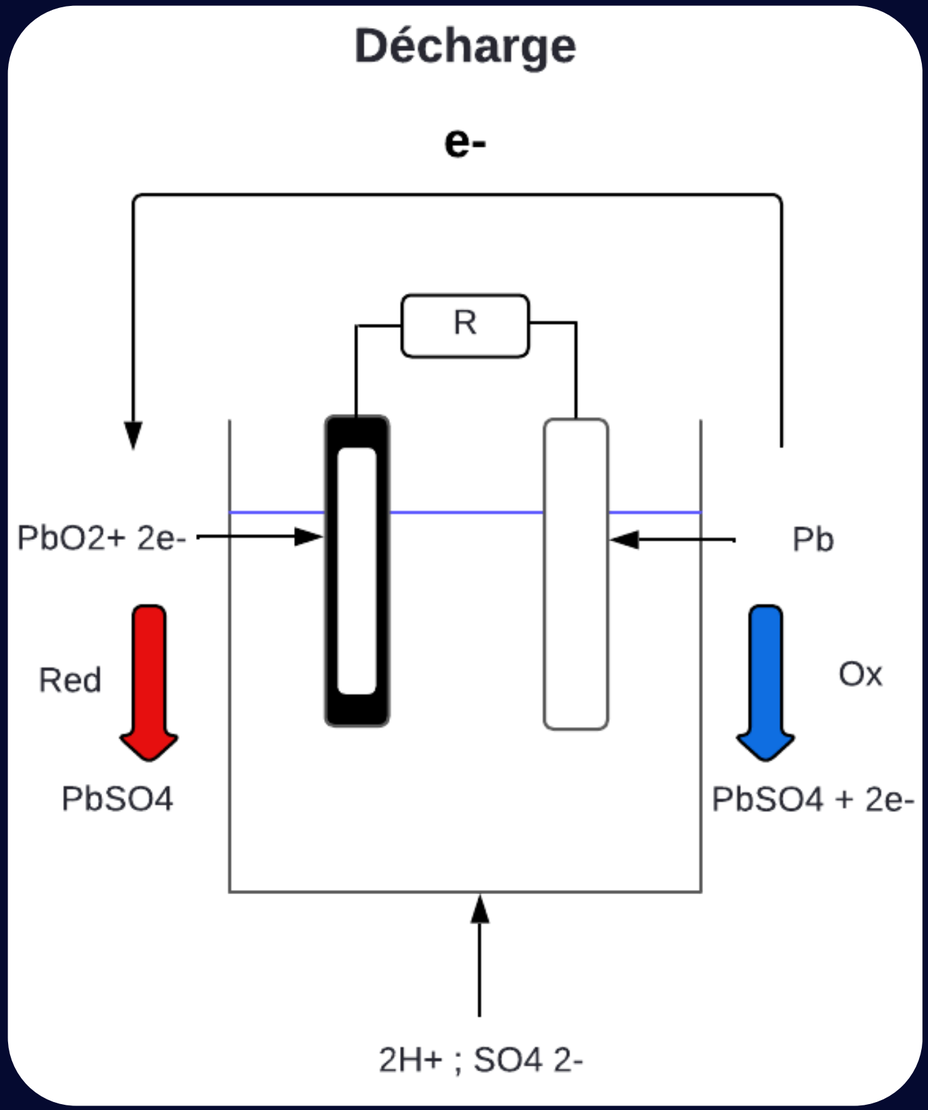


CHARGE

$PbSO_4 + 2H_2O \rightarrow Pb + PbO_2 + 2H_2SO_4$

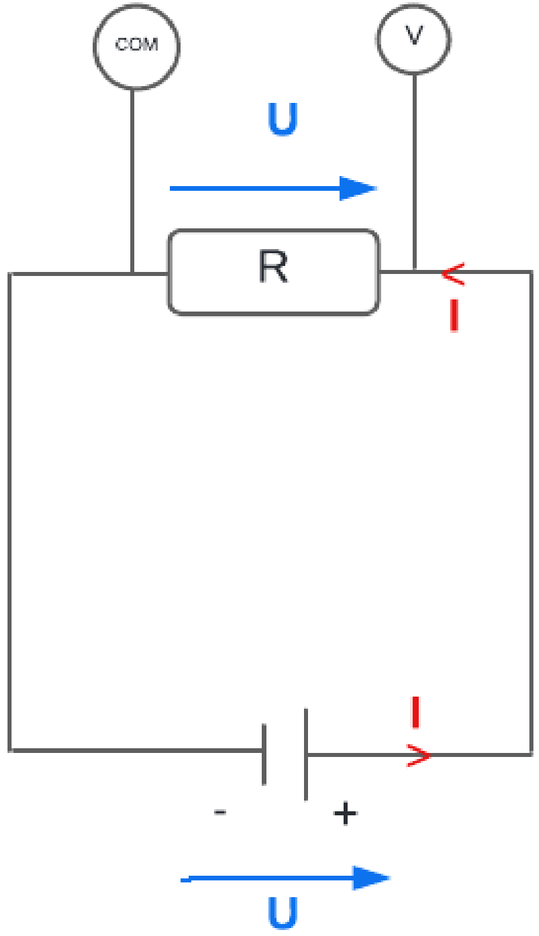
DÉCHARGE

$Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$

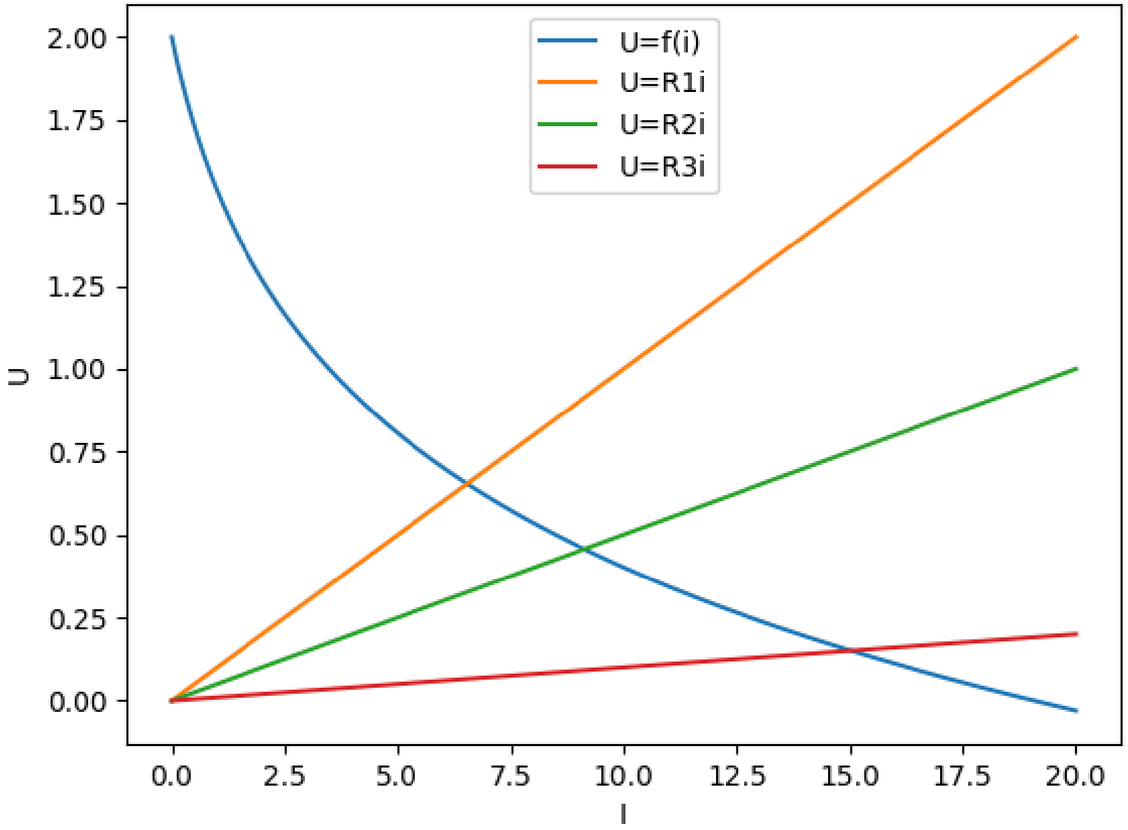


1 MODÈLE CARACTÉRISTIQUE

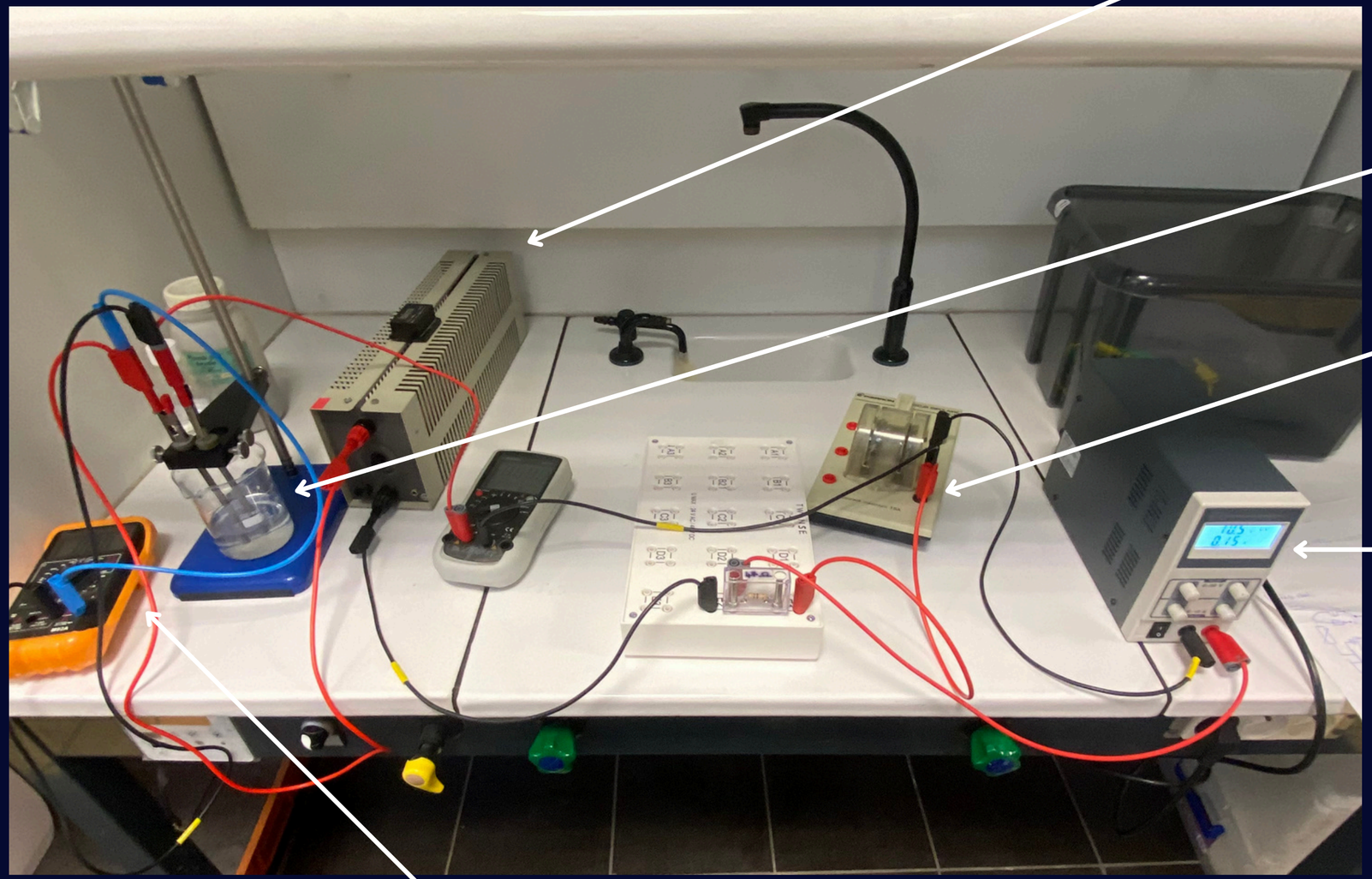
MONTAGE



PRINCIPE



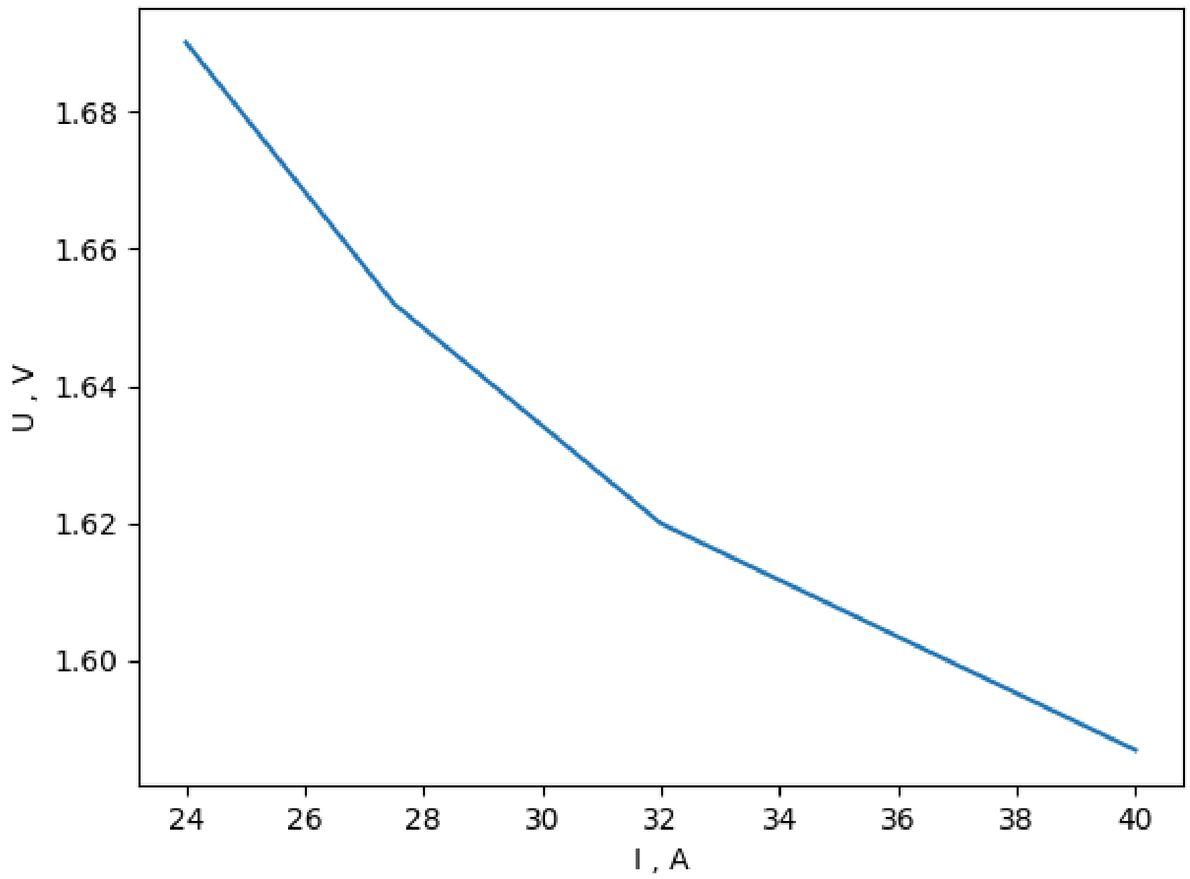
1 MODÈLE


RÉSISTANCE VARIABLE
PILE
INTERRUPTEUR
GÉNÉRATEUR
VOLTMÈTRE

1 MODÈLE

CARACTÉRISTIQUE

COURBE EXPÉRIMENTALE



1 MODÈLE

INTERPRÉTATION

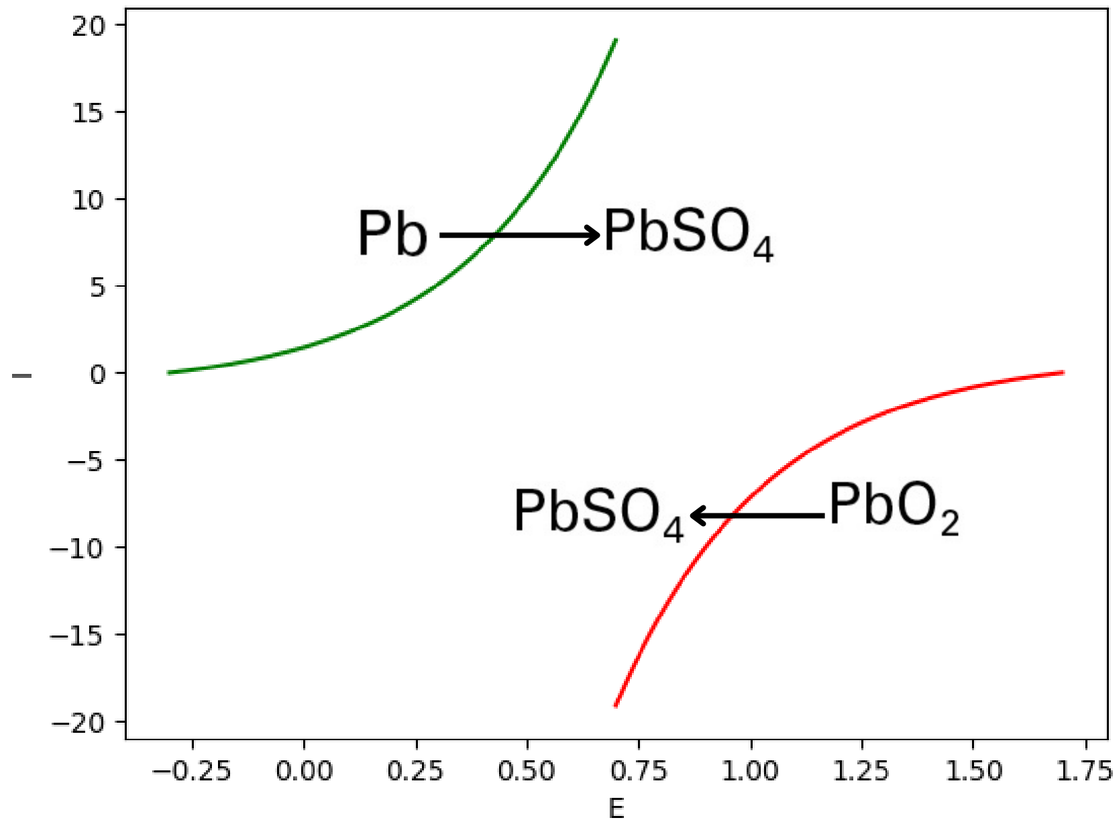
Anode

I(E)=exp(E-Eéq1)-1

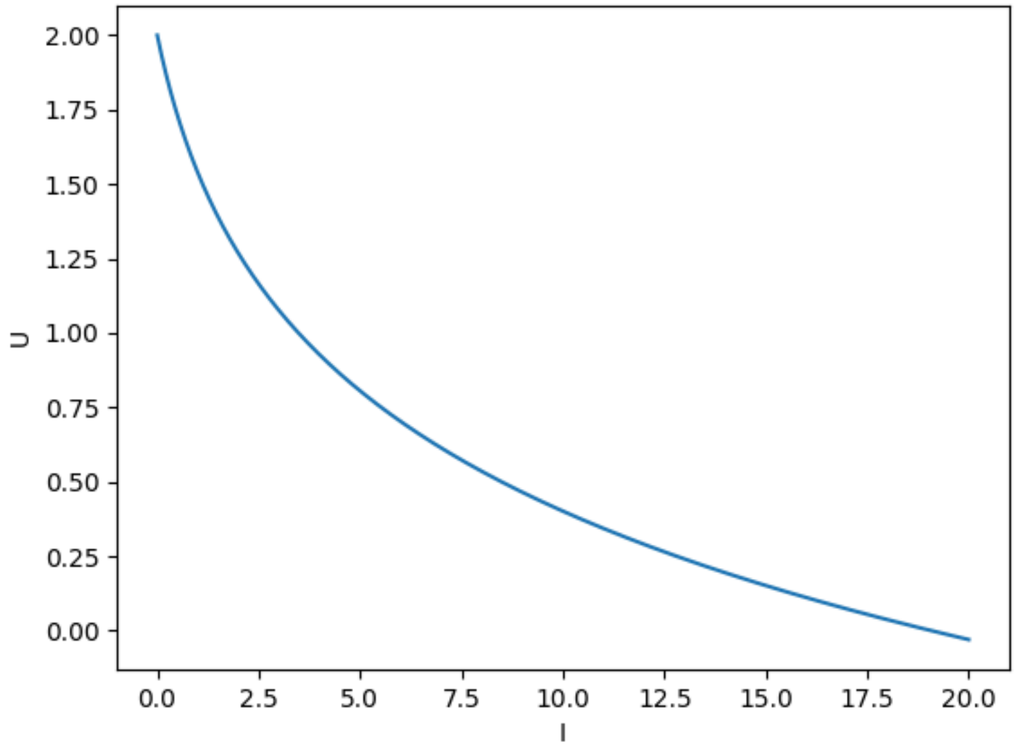
Cathode

I(E)=1-exp(-)(E-Eéq2)

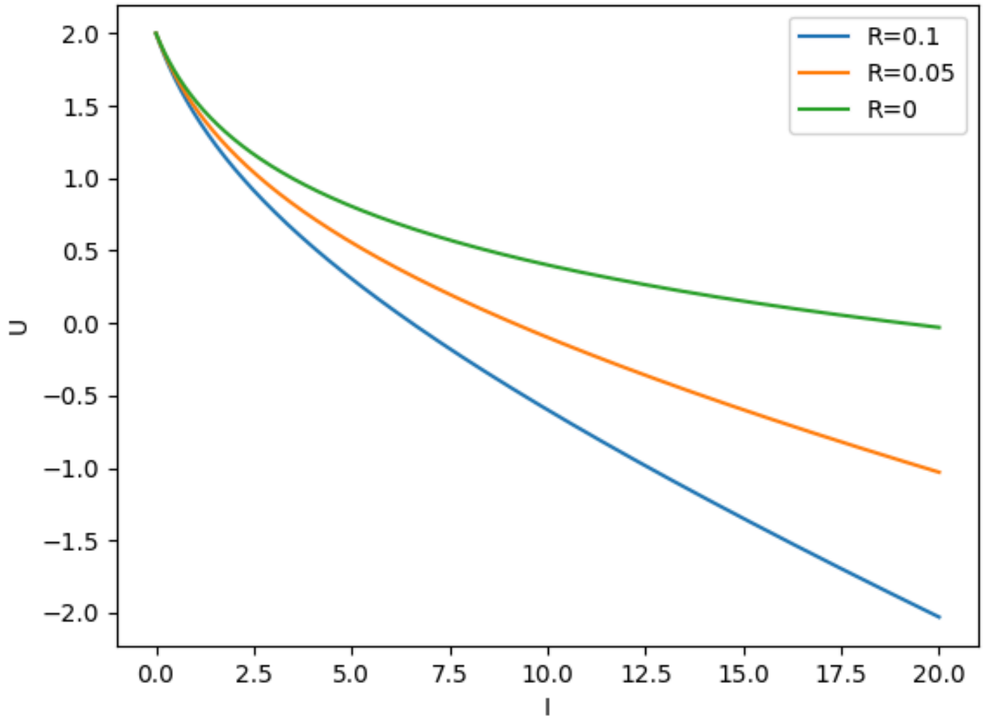
CARACTÉRISTIQUE



SANS RÉSISTANCE INTERNE

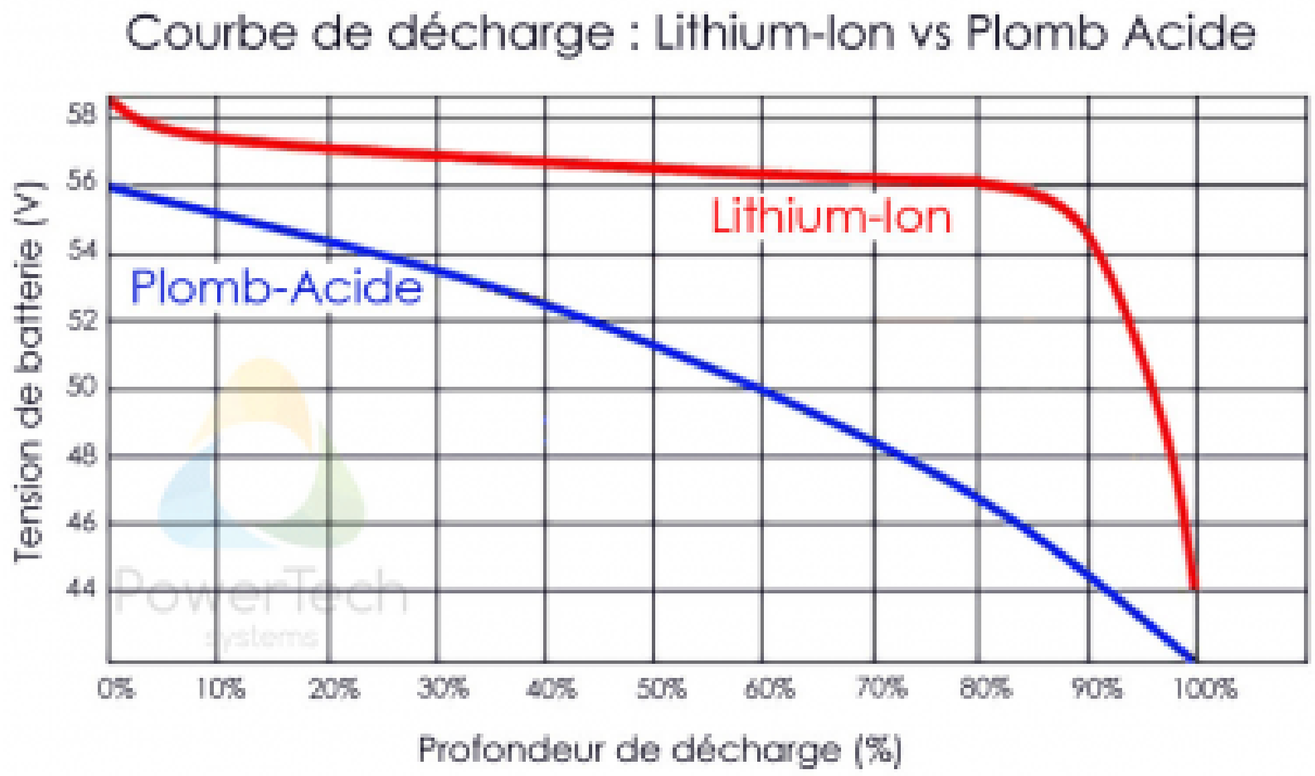


AVEC RÉSISTANCE INTERNE



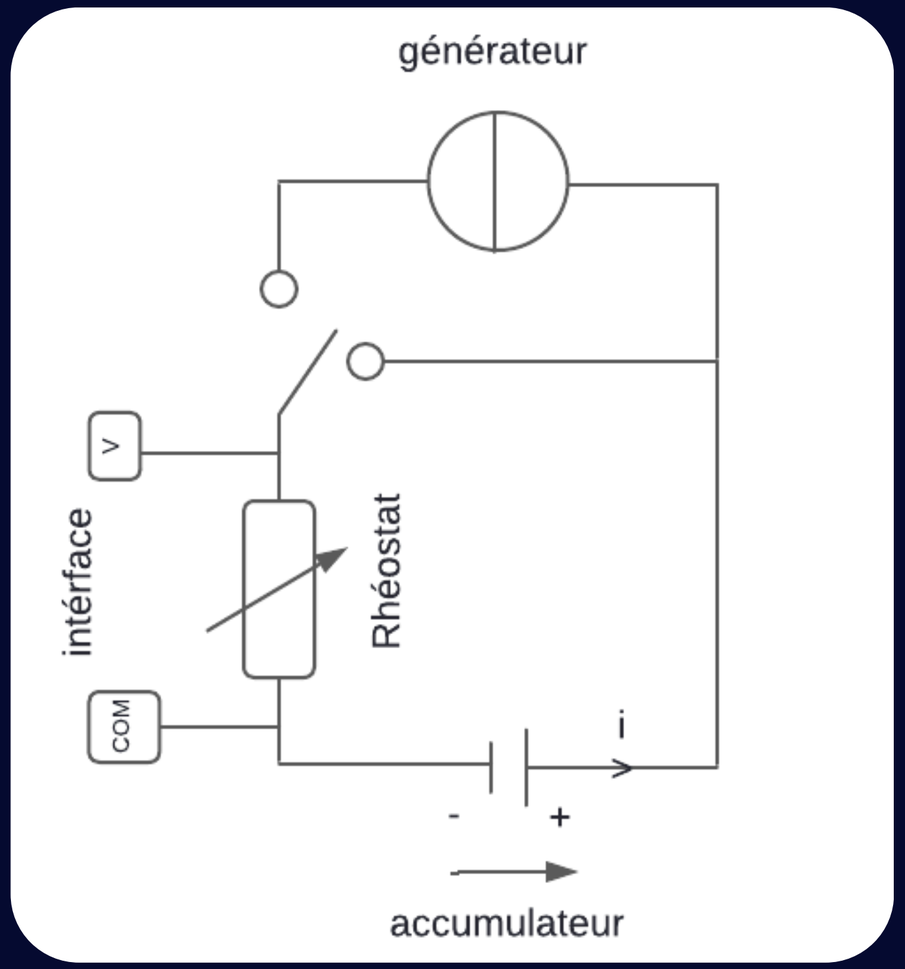
2 INCONVÉNIENT:
INFLUENCE DE LA PUISSANCE DE DÉCHARGE

BATTERIE LITHIUM-ION / PLOMB-ACIDE



2 INFLUENCE DE L'INTENSITÉ

MONTAGE



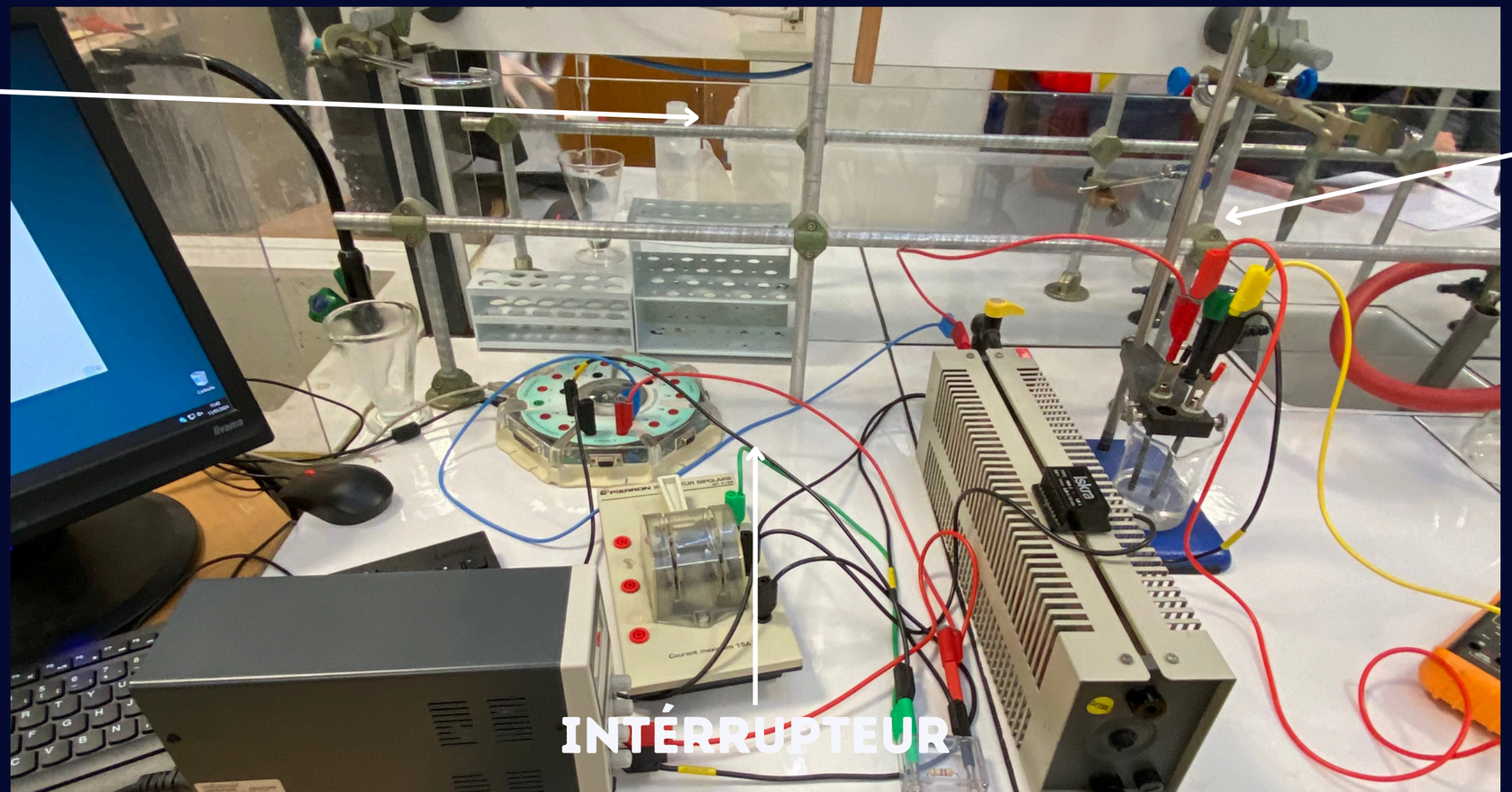
PRINCIPE

U=RI

R VARIE

I ET U VARIENT

2 INFLUENCE DE L'INTENSITÉ



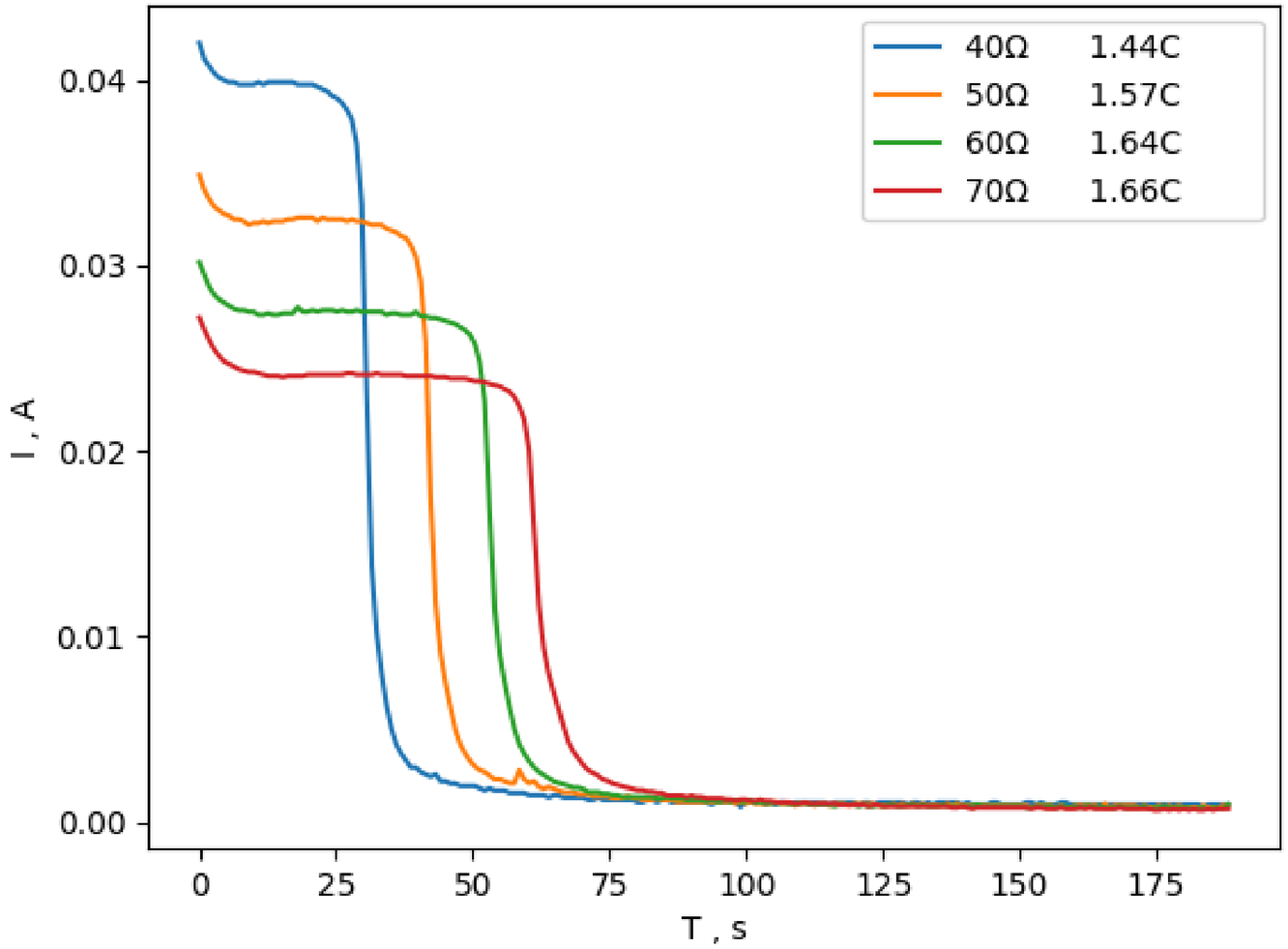
2 INFLUENCE DE L'INTENSITÉ

COURBE EXPÉRIMENTALE

COURANT DE DÉCHARGE ↑

↓

RENDEMENT ↓



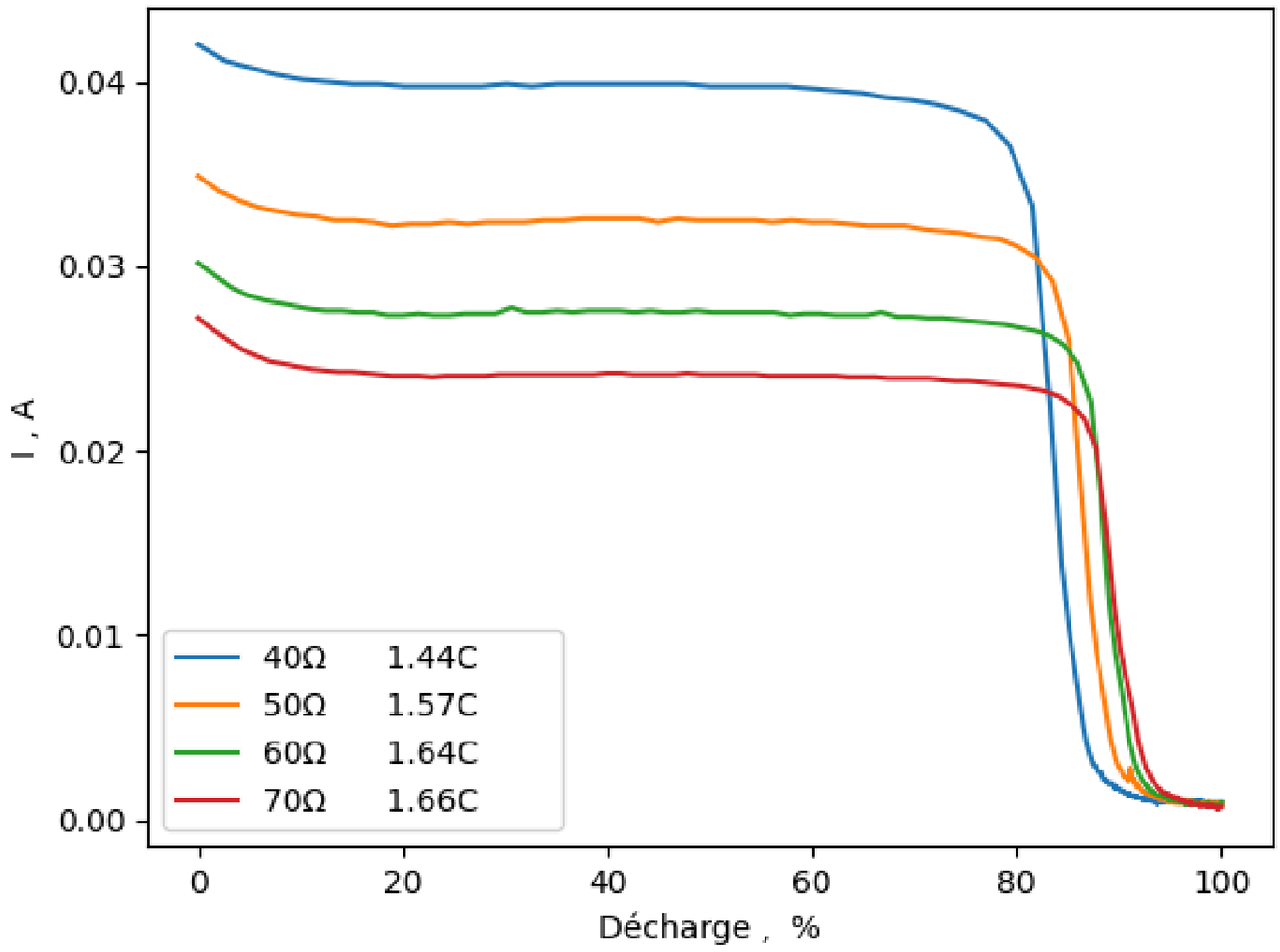
2 INFLUENCE DE L'INTENSITÉ

COURBE EXPÉRIMENTALE

COURANT DE DÉCHARGE ↑

↓

CHUTE DE L'INTENSITÉ ↑



2 INFLUENCE DE L'INTENSITÉ

RÉSISTANCE INTERNE

MODÈLE

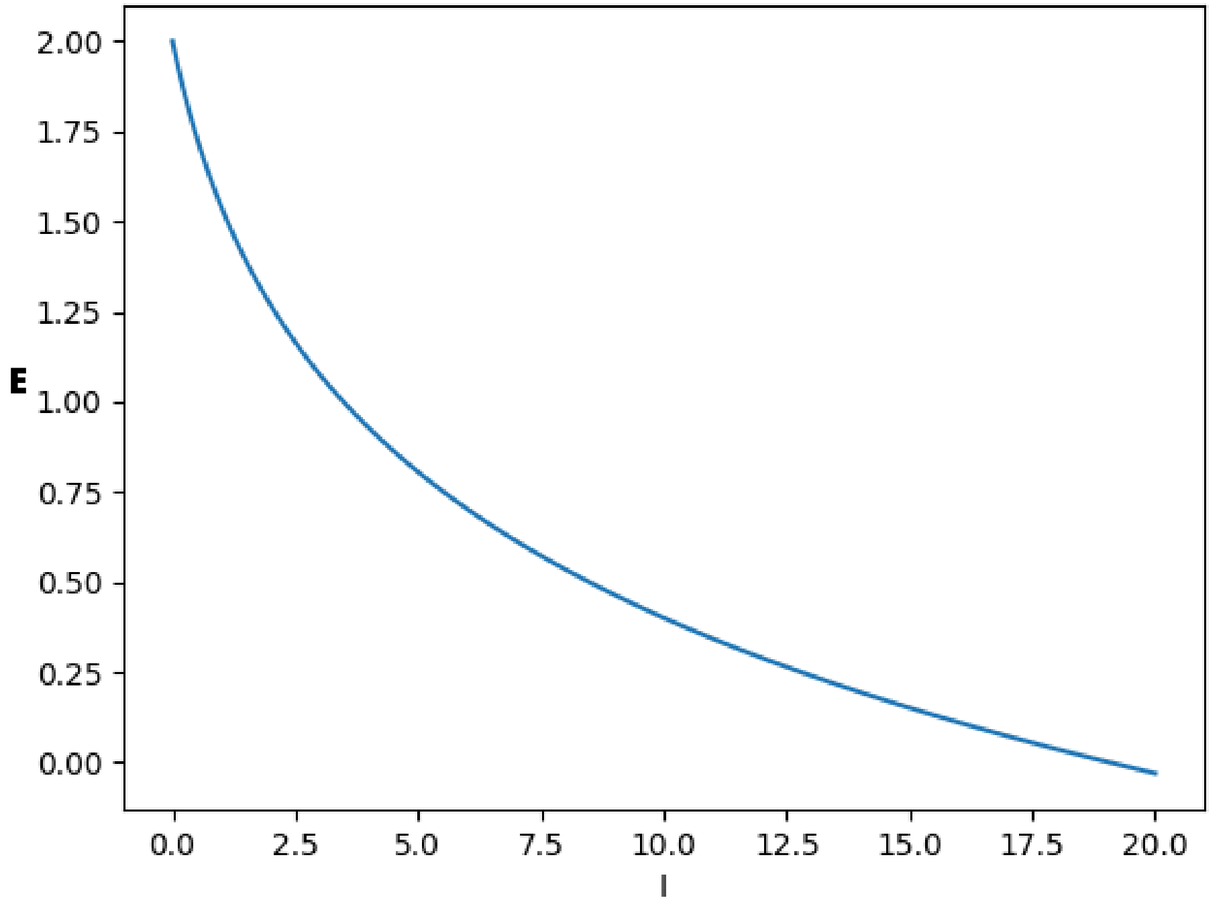
U(I)= E-RI

VALABLE SI:

E VARIE PAS TROP

⬇

I GRAND



2 INFLUENCE DE L'INTENSITÉ

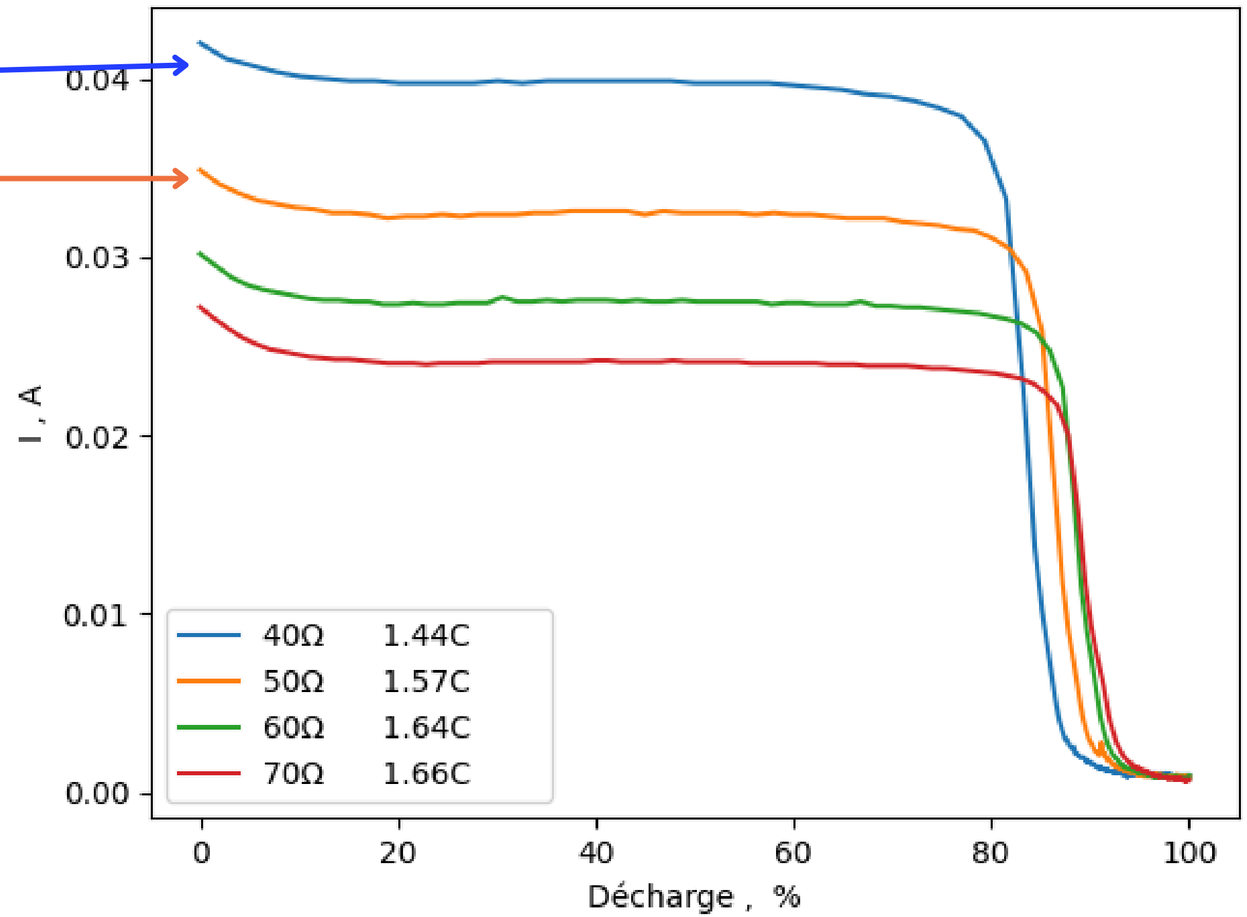
RÉSISTANCE INTERNE 20C°

U(I40)= E-RI40

U(I50)= E-RI50

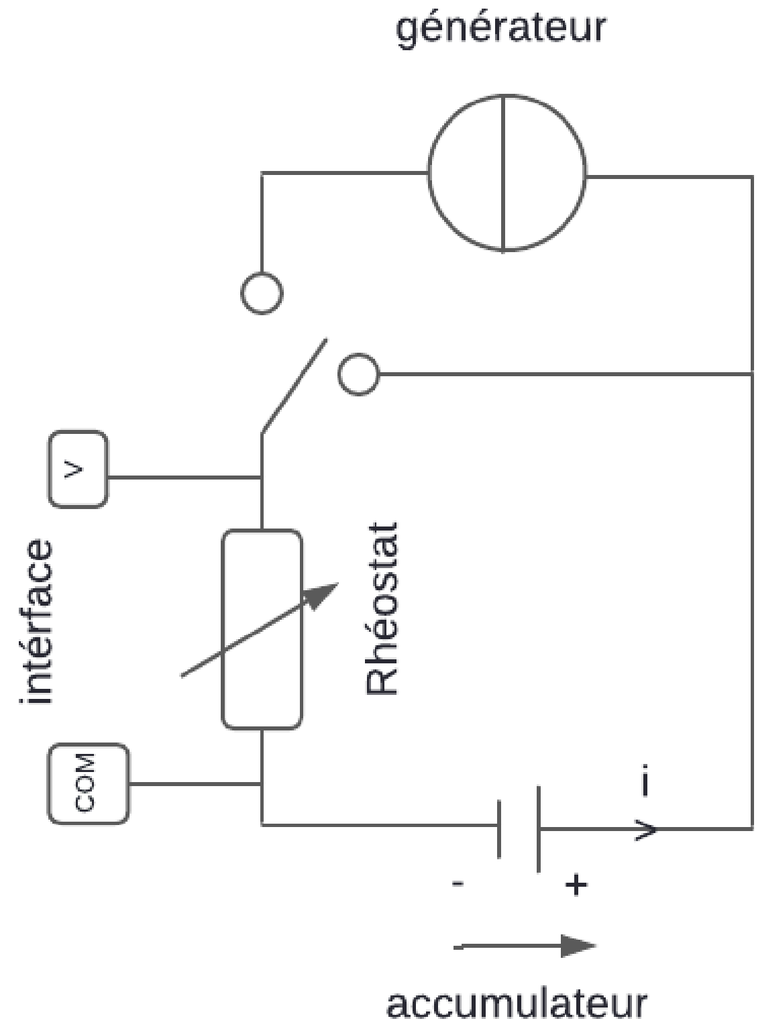
$$R = \frac{U(I50)-U(I40)}{I40-I50}$$

R=5,1Ω



3 INFLUENCE DE LA TEMPÉRATURE

MONTAGE



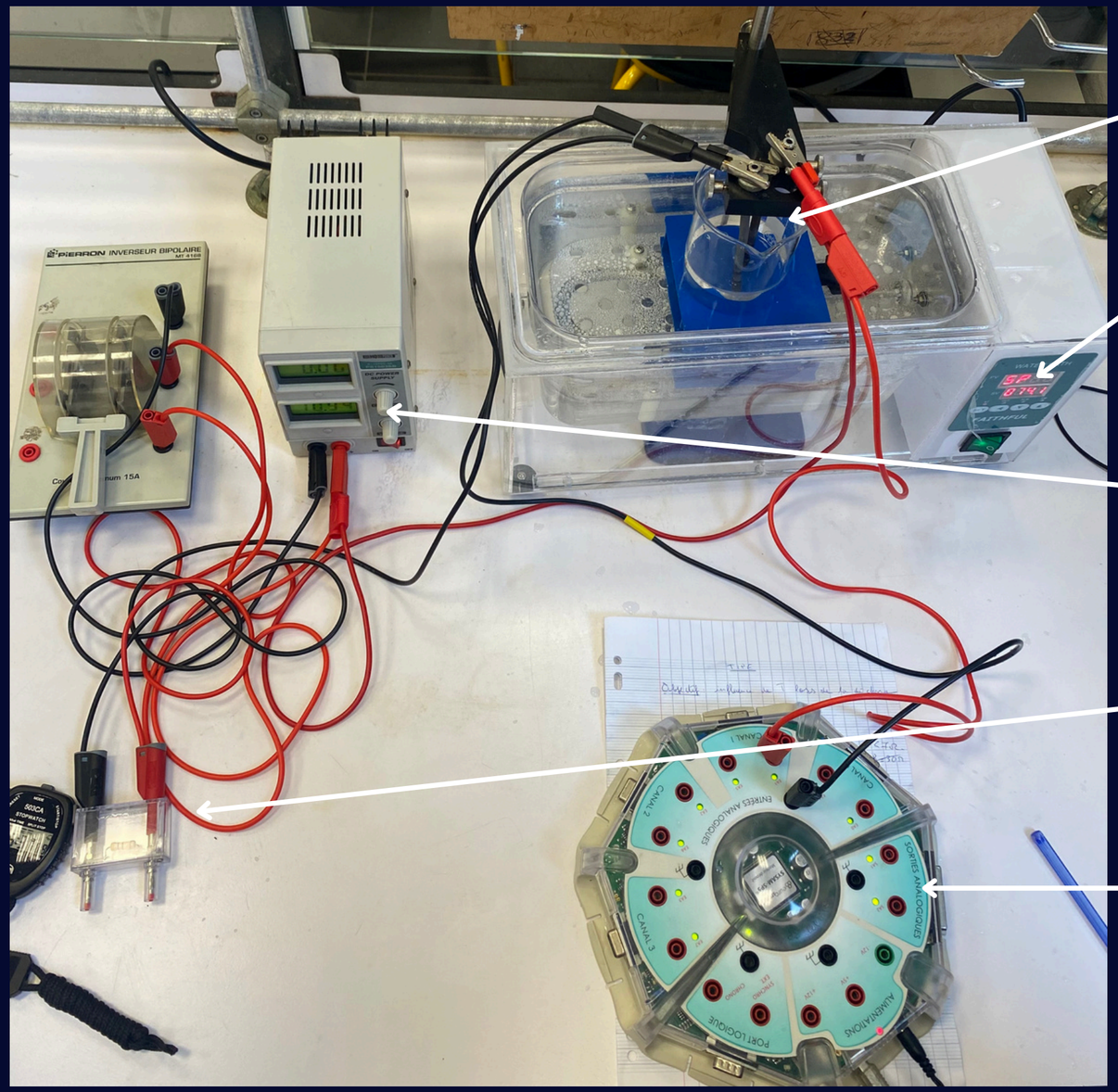
PROTOCOLE

DÉCHARGES À T° DIFFÉRENTES

- **CHARGES: IDENTIQUES PENDANT 8MIN À 150 MA**

- **DÉCHARGES À 10C° ,20C°,40C°À R=70Ω**

3 INFLUENCE DE LA TEMPÉRATURE



PILE

BAIN THERMOSTATÉ

GÉNÉRATEUR

RESISTANCE

INTERFACE

MONTAGE

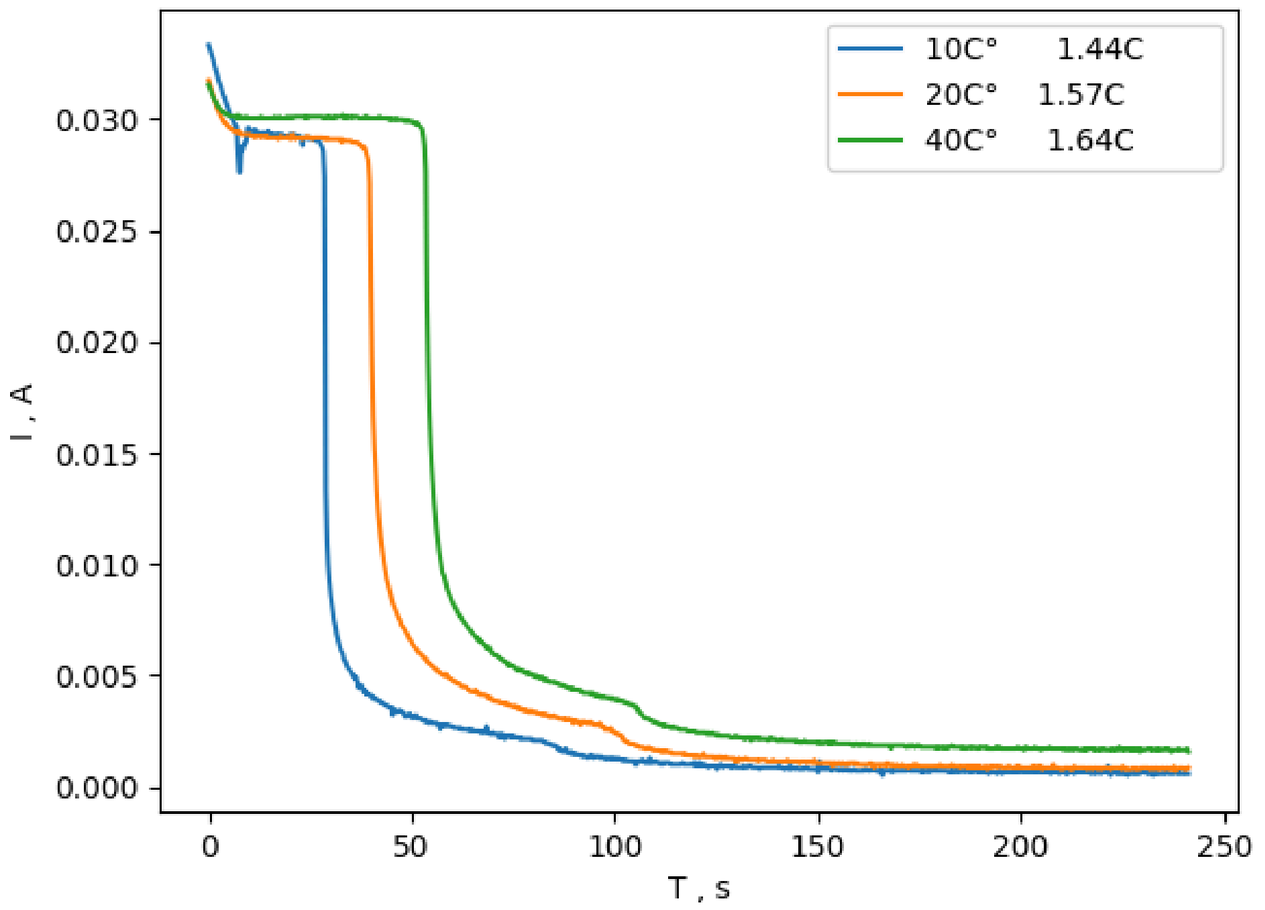
3 INFLUENCE DE LA TEMPÉRATURE

COURBE EXPÉRIMENTALE

TEMPÉRATURE ↗

↓

RENDEMENT ↗



3 INFLUENCE DE LA TEMPÉRATURE

RÉSISTANCE INTERNE

MODÈLE : U = E - RI **E NE VARIE PAS**

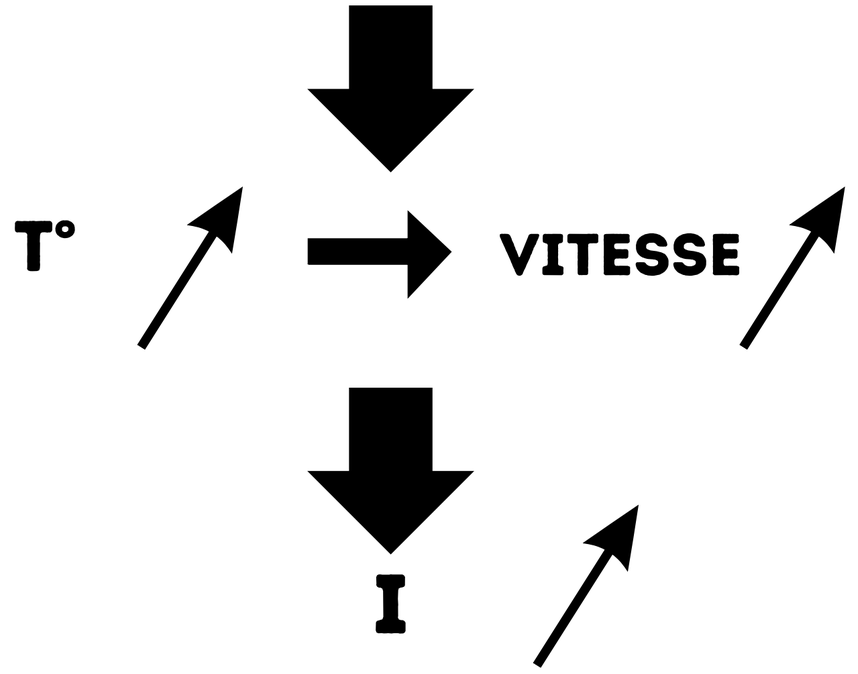
$$\begin{cases} U_{10} = E - R_{10}I_{10} \\ U_{20} = E - R_{20}I_{20} \\ U_{40} = E - R_{40}I_{40} \end{cases}$$

R(20 C°)=5,1 Ω

CINÉTIQUE

LOI D'ARRHÉNIUS

$$k = A \cdot \mathrm{e}^{\frac{-E_\mathrm{a}}{RT}}$$



3 INFLUENCE DE LA TEMPÉRATURE

THERMODYNAMIQUE

$U = E2-E1 + \eta_{2}^{(i)} - \eta_{1}^{(i)} - Ri$

- THERMO
- CINÉTIQUE
- CHUTE OHMIQUE

LOI DE HESS $\Delta_r H^\circ$ **= -315.86** kJ mol^{-1} **< 0**

REACTION EXOTHERMIQUE

T° ↗ → **RENDEMENT** ↘

ANNEXE

PRINCIPE CARACTÉRISTIQUE (5)

```
import numpy as np
from math import *
import matplotlib.pyplot as plt
import math as m
dx=0.0001
def a(x,r):
    return r*x
def u(i,r):
    return 2-2*m.log(1+i)/3-r*i
def l(f,a,b,r):
    X=[]
    Y=[]
    c=a
    while c<b:
        X.append(c)
        Y.append(f(c,r))
        c+=dx
    return X,Y
Xa,Ya=l(u,0,20,0)
Xb,Yb=l(a,0,20,0.1)
Xc,Yc=l(a,0,20,0.05)
Xd,Yd=l(a,0,20,0.01)
plt.plot(Xa,Ya,label='U=f(i)')
plt.plot(Xb,Yb,label='U=R1i')
plt.plot(Xc,Yc,label='U=R2i')
plt.plot(Xd,Yd,label='U=R3i')
plt.legend()
plt.xlabel('I')
plt.ylabel('U')
plt.show()
```

COURBE EXPÉRIMENTALE CARACTÉRISTIQUE (7)

```
import math as m
import matplotlib.pyplot as plt
I=[40,32,27.5,24]
U=[1.587,1.62,1.652,1.69]
plt.plot(I,U)
plt.xlabel('I , A')
plt.ylabel('U , V')
plt.show()
```

COURBE THÉORIQUE CARACTÉRISTIQUE (9)

```
import numpy as np
from math import *
import matplotlib.pyplot as plt
import math as m
dx=0.0001
def a(x):
    return exp(3*(x+0.3))-1
def c(x):
    return 1-exp(-3*(x-1.7))
def u(i):
    return 2-2*m.log(1+i)/3
def l(f,a,b):
    X=[]
    Y=[]
    c=a
    while c<b:
        X.append(c)
        Y.append(f(c))
        c+=dx
    return X,Y
Xa,Ya=l(a,-0.3,0.7)
Xb,Yb=l(c,0.7,1.7)
I,U=l(u,0,18)
plt.plot(I,U,'y')
plt.xlabel('I')
plt.ylabel('U')
plt.show()
```

INTENSITÉ EN FONCTION DU TEMPS PUISSANCE (13)

```
import csv
import matplotlib.pyplot as plt
def integrale(T1,X1,X2,X3,X4):
    dt=T1[3]-T1[2]
    C1,C2,C3,C4=0,0,0,0
    for i in range (len(X1)-1):
        C1=C1+(X1[i]+X1[i+1])*dt/2
    for i in range (len(X2)-1):
        C2=C2+(X2[i]+X2[i+1])*dt/2
    for i in range (len(X3)-1):
        C3=C3+(X3[i]+X3[i+1])*dt/2
    for i in range (len(X4)-1):
        C4=C4+(X4[i]+X4[i+1])*dt/2
    return(C1,C2,C3,C4)
def courbes(a,b,c,d):
    T1=[0]
    X1=[0]
    T2=[0]
    X2=[0]
    T3=[0]
    X3=[0]
    T4=[0]
    X4=[0]
    with open(a, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T1.append(float(row[0]))
            X1.append(float(row[1]))
        X1.remove(X1[0])
        T1.remove(T1[0])
    with open(b, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T2.append(float(row[0]))
            X2.append(float(row[1]))
        X2.remove(X2[0])
        T2.remove(T2[0])
    with open(c, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T3.append(float(row[0]))
            X3.append(float(row[1]))
        X3.remove(X3[0])
        T3.remove(T3[0])
    with open(d, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T4.append(float(row[0]))
            X4.append(float(row[1]))
        X4.remove(X4[0])
        T4.remove(T4[0])
    return(T1,T2,T3,T4,X1,X2,X3,X4)
T1,T2,T3,T4,X1,X2,X3,X4=courbes('...','...','...','...')
print(integrale(T1,X1,X2,X3,X4))
plt.plot(T1,X1,label='40Ω      1.44C       ')
plt.plot(T2,X2,label='50Ω      1.57C       ')
plt.plot(T3,X3,label='60Ω      1.64C       ')
plt.plot(T4,X4,label='70Ω      1.66C       ')
plt.legend()44
plt.ylabel('I , A')
plt.xlabel('T , s')
plt.show()
```

INTENSITÉ EN FONCTON DU TAUX DE DÉCHARGE PUISSANCE (14)

```
import csv
import matplotlib.pyplot as plt
def integrale(T1,X1):
    dt=0.9
    C1=0
    for i in range (len(X1)-1):
        C1=C1+(X1[i]+X1[i+1])*dt/2
    return(C1)
C3=[]
C1=[]
C2=[]
C4=[]
def courbes(a,b,c,d):
    T1=[0]
    X1=[0]
    T2=[0]
    X2=[0]
    T3=[0]
    X3=[0]
    T4=[0]
    X4=[0]
    with open(a, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T1.append(float(row[0]))
            X1.append(float(row[1]))
        X1.remove(X1[0])
        T1.remove(T1[0])
    with open(b, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T2.append(float(row[0]))
            X2.append(float(row[1]))
        X2.remove(X2[0])
        T2.remove(T2[0])
    with open(c, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T3.append(float(row[0]))
            X3.append(float(row[1]))
        X3.remove(X3[0])
        T3.remove(T3[0])
    with open(d, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T4.append(float(row[0]))
            X4.append(float(row[1]))
        X4.remove(X4[0])
        T4.remove(T4[0])
    return(T1,T2,T3,T4,X1,X2,X3,X4)
T1,T2,T3,T4,X1,X2,X3,X4=courbes('...','...','...','...')
for i in range (1,len(T1)+1):
    C1.append(100*integrale(T1[0:i],X1[0:i])/1.4408641151717165)
    C2.append(100*integrale(T2[0:i],X2[0:i])/1.5688467946657938)
    C3.append(100*integrale(T3[0:i],X3[0:i])/1.6393281710206027)
    C4.append(100*integrale(T4[0:i],X4[0:i])/1.6536124282632993)
plt.plot(C1,X1,label='40Ω      1.44C       ')
plt.plot(C2,X2,label='50Ω      1.57C       ')
plt.plot(C3,X3,label='60Ω      1.64C       ')
plt.plot(C4,X4,label='70Ω      1.66C       ')
plt.legend()
plt.ylabel('I , A')
```

INTENSITÉ EN FONCTION DU TEMPS TEMPÉRATURE (19)

```
import csv
import matplotlib.pyplot as plt
def integrale(T,X):
    dt=0.3
    C=0
    for i in range (len(X)):
        C=C+X[i]*dt
    return(C)
def courbes(a,b,c):
    T1=[0]
    X1=[0]
    T2=[0]
    X2=[0]
    T3=[0]
    X3=[0]
    with open(a, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T1.append(float(row[0]))
            X1.append(float(row[1])/70)
        X1.remove(X1[1])
        T1.remove(T1[1])
    with open(b, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T2.append(float(row[0]))
            X2.append(float(row[1])/70)
        X2.remove(X2[1])
        T2.remove(T2[1])
    with open(c, 'r', newline='', encoding='utf-8') as csvfile:
        reader = csv.reader(csvfile, delimiter=';')
        header = next(reader)
        for row in reader:
            T3.append(float(row[0]))
            X3.append(float(row[1])/70)
        X3.remove(X3[1])
        T3.remove(T3[1])
    return(T1,T2,T3,X1,X2,X3)
T1,T2,T3,X1,X2,X3=courbes('/Users/anastariq/Desktop/TIPE/Temperature 70ohm/E10c2.
                          '/Users/anastariq/Desktop/TIPE/Temperature 70ohm/E20c2.
                          '/Users/anastariq/Desktop/TIPE/Temperature 70ohm/E40c2.
C1=[]
C2=[]
C3=[]
for i in range (1,len(T1)+1):
    C1.append(100*integrale(T1[0:i],X1[0:i]))
    C2.append(100*integrale(T2[0:i],X2[0:i]))
    C3.append(100*integrale(T3[0:i],X3[0:i]))
print(integrale(T1,X1),integrale(T2,X2),integrale(T3,X3))
plt.plot(C1,X1,label='10C°      1.44C       ')
plt.plot(C2,X2,label='20C°    1.57C       ')
plt.plot(C3,X3,label='40C°     1.64C       ')
plt.legend()
plt.ylabel('I , A')
plt.xlabel('Décharge ,  %')
plt.show()
```